

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

22nd May 2012

Subject CA1 – Actuarial Risk Management (Paper II)

Time allowed: 3 Hours (14.45* - 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
2. *Attempt all questions, beginning your answer to each question on a separate sheet.*
3. **You have 15 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have 3 hours to complete the paper.*
4. *You must not start writing your answers until instructed to do so by the Supervisor.*
5. *Mark allocations are shown in brackets.*
6. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** In the context of measurement of investment risk explain each of the following measures with an example
- i)** VaR (2)
 - ii)** Tail VaR (2)
- [4]**
- Q. 2)** A life insurance company writes traditional with and without profits endowment, term insurance and immediate annuity products. The company offers discontinuance terms for some products.
- i)** What do you understand by discontinuance in the context of life insurance products? (2)
 - ii)** Discuss for which of the above products the company may not provide any lump sum discontinuance terms and why? (5)
- [7]**
- Q. 3)**
- i)** What are the characteristics of commercial property insurance and travel insurance in terms of
 - (a)** Purpose of the cover
 - (b)** Benefits
 - (c)** Perils covered (6)
 - ii)** List the factors that need to be considered when pricing a travel insurance policy. (3)
- [9]**
- Q. 4)**
- i)** State the variations in the level and form of the benefits that may be made available in term insurance contracts. (3)
- A life insurance company sells a term insurance product with an inbuilt critical illness benefit. The benefits available under the product are as follows-
- On being diagnosed with any of the defined critical illnesses during the policy term, 50% of the Sum assured is paid to the customer. The balance 50% of the Sum Assured is payable on subsequent death if it occurs during the policy term
 - Otherwise, on death of policyholder during the term of the policy 100% Sum Assured is payable.
- ii)** Discuss the principal sources of profits/losses on a portfolio of above policies. (6)
- [9]**

- Q. 5)** A municipal body of a metropolitan city has put up for auction a 15 year lease on a property adjacent to a local railway station. This property is to be developed into a multi storied parking lot for cars and two-wheelers. The property will revert to the municipal body at the end of lease period of 15 years. A property developer proposes to bid for this auction.
- i)** Discuss the steps that need to be taken by the developer to identify the risks in this project before making a formal bid. (6)
 - ii)** State the major risks in the project and how these can be mitigated? (6)
- [12]**
- Q. 6)**
- i)** State the main reasons for a general insurance company to consider reinsurance? (5)
 - ii)** Describe, why general insurance companies may use alternative risk transfer (ART) contracts rather than traditional reinsurance to manage risk?
- A general insurance company sells different types of liability covers primarily employer's liability insurance. (6)
- iii)** Describe briefly why the company may look at purchasing discounted covers. (4)
- [15]**
- Q. 7)** An employer of a small company wants to set up a defined contribution pension scheme for his employees.
- i)** Discuss why an employer may provide pension benefits to his employees.
- As a Consultant Actuary, the employer has sought your recommendation on the contribution rate payable by both the employer and employee to the new scheme. (4)
- ii)** Discuss the factors to consider before making your recommendation. (8)
 - iii)** Describe briefly the main economic assumptions you would need to consider in determining the contribution rates. (4)
- [16]**

Q. 8) A life insurance company wants to enter into health insurance business for the first time. The product development team has proposed the following design for this new health product to the Senior Management Team.

- A fixed benefit per day is payable for minimum 24 continuous hours stay in hospital as daily hospital benefit.
- A fixed benefit per day is payable for minimum 24 continuous hours stay in ICU, which is twice the fixed daily hospital benefit.
- A fixed benefit in lump sum varying in scale from 10 times to 100 times of daily hospital cash benefit, is payable for undergoing a listed surgical procedure, with the scale depending on the grade of the surgery subject to minimum 24 hours hospitalization for surgery except for Day Care surgeries.
- Listed surgical procedure means any procedure listed under the terms and conditions of the policy.
- No surrender, death and maturity benefits are available.
- The policy is of regular premium and premiums are guaranteed for 5 years and are reviewable after every 5 years during the term of the policy.
- There is a benefit of No Claim Bonus (NCB) if the policyholder makes no claim during a policy year.

i) Discuss the factors that the Senior Management team needs to consider while approving the design for the health insurance product? (13)

ii) Explain why the company should consider offering a No Claim Bonus benefit. (2)

iii) Suggest a benefit structure for the No Claim Bonus (NCB). (3)

The Marketing Head of the company has suggested that instead of NCB, a discount (called No Claim Discount) be offered on premium payable in the succeeding policy years.

iv) Suggest a structure for the No Claim Discount (NCD). (2)

v) Discuss the factors that need to be considered when deciding whether to offer NCB or NCD for this product. (8)

[28]
